Muhammad Abdullah Goher

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Education

University of Pennsylvania — Aug 2023 – May 2027
B.S.E. & M.S.E. in Computer Science (Submatriculation), Concentration in Artificial Intelligence — *Philadelphia, PA*
GPA: 3.71 (B.S.E.) / 3.85 (M.S.E.) | **Coursework:** Applied Machine Learning, Artificial Intelligence, Big Data Analytics, Probability, Statistical Inference, Linear Algebra, Algorithms, Databases

Experience

AeternalLabs, PBC – Founding Machine Learning Engineer — May 2026 – Present
Runtime AI-governance platform measuring demographic bias in LLM outputs (first engineering hire) — *Remote*

- Built the ML evaluation infrastructure for a counterfactual-fairness system spanning **8 providers and 18+ frontier models** (OpenAI, Anthropic, Google, xAI, DeepSeek, Cerebras) behind a FastAPI service.
- Implemented a 22-dimension asymmetry scoring vector with statistical comparison tests to flag demographic disparities in model responses.
- Parallelized counterfactual A/B evaluation runs with a thread pool, cutting runtime **1.88**× (158s → 84s).
- Designed a PostgreSQL store (SQLAlchemy) logging every model run with integrity hashing for reproducible, auditable experiments.

PennAdapt – Machine Learning Engineer — Sep 2024 – Present
Penn Assistive Devices & Prosthetic Technologies — *Philadelphia, PA*

- Trained a real-time computer-vision model (CNN + Vision Transformer, PyTorch) detecting **26 classes** of surgical tools at **90%+** accuracy across varying angles.
- Curated and annotated a **1,500+** image dataset with Penn Medicine surgeons; ran iterative training and hyperparameter optimization within a 10-engineer team.

Computational Social Science Lab (Prof. Duncan Watts) – Machine Learning Engineer — May 2025 – Dec 2025
LLM pipelines & large-scale geospatial modeling — *Philadelphia, PA*

- Built an LLM-powered pipeline extracting structured features from **100+** scholarly papers, plus a RAG layer surfacing quantitative mobility metrics.
- Modeled crowd density from multi-terabyte GPS datasets (Meta, Factori, SafeGraph) using PySpark on AWS EMR for 2026 FIFA World Cup planning.

Clab AI – Machine Learning Intern — May 2024 – Aug 2024
AI-powered college-application assistant used by 100+ students — *Hybrid – Nashville, TN*

- Built hyperparameter-tuned Random Forest / Linear Regression models predicting financial-aid eligibility at **82%** R^2, with EDA across 100+ universities.
- Fine-tuned LLMs on hundreds of accepted essays using RLHF to personalize writing guidance.

Projects

News Source Classification | *PyTorch, HuggingFace Transformers, scikit-learn*

- Co-built (team of 3) a headline classifier distinguishing Fox vs. NBC; scaled the dataset **3,800** → **17,900** headlines via multithreaded sitemap and Apify scraping pipelines.
- Fine-tuned RoBERTa/DeBERTa backbones (AdamW, cosine schedule, mixed precision); best model (RoBERTa-base + LinearSVC head on [CLS] embeddings) reached **91.75% accuracy**, +25 pts over a TF-IDF baseline.

Steam Dataset EDA & Modeling | *Python, scikit-learn, XGBoost, LightGBM*

- Cleaned and modeled a **110k+** game dataset; trained Random Forest, XGBoost, and LightGBM ensembles to predict commercial success and surface feature importance.

Technical Skills

Machine Learning: PyTorch, TensorFlow/Keras, scikit-learn, HuggingFace Transformers, CNNs, Vision Transformers, NLP, XGBoost/LightGBM, fine-tuning (LoRA/PEFT, RLHF), RAG
Languages: Python, C, C++, Java, JavaScript/TypeScript, SQL
Data & Stats: pandas, NumPy, PySpark, hypothesis testing, feature engineering, Matplotlib, Jupyter, vector DBs (FAISS/Chroma)
Backend & Infra: FastAPI, Node.js, Express, PostgreSQL, MongoDB, Docker, AWS (S3, EMR), Git, Linux